

To boost the neural network's performance, jittering is used for data augmentation. Basically, I randomly shifted and rotated each training image during the training process. I'd move the image up, down, left, right, and spin it by 15 degrees in both directions. To do this, I created a new method which utilized `np.roll` and `rotate`. The jittered network gives an advantage in both training and testing accuracy. The number of misclassified data points per epoch down, which means the model was better at generalizing overall. The jittered network seemed to generalize better, with higher testing accuracy and fewer misclassified points. This suggests that introducing those random transformations during training helps the network learn more invariant features, leading to better results.

Blue MNIST accuracy::

Misclassified for `mnist_train.csv`: 0.212%

Misclassified for `mnist_test.csv`: 1.87%

Distortions Red MNIST accuracy

Misclassified for `mnist_train.csv`: 0.016%

Misclassified for `mnist_test.csv`: 0.73%